Causal statistics for treatment models

6. Design-based inference

Ecole Normale Supérieure–PSL, L3 Economics and Cogmaster, 2025

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Today

We will talk about inference. Inference refers to the fact that you infer something about the true parameter: test a null on its value with a t-test, compute confidence intervals, etc. Remember that **the central piece in all these actions is the variance of the estimator** (and its asymptotic distribution, that is the normal distribution).

So, we will wonder *what exactly* makes the estimator a random variable (thus having a variance and a normal distribution), and look closely into the ways we should compute the variances.

Starting example: average age

a_i is the age of each one in this class

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is the average age in the class

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Is τ a random variable? Does it have a mean $E(\tau)$, a variance $V(\tau)$?

Starting example: average age

If the object of interest is average age IN THIS CLASS

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is a parameter and its value is estimated with certainty

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If the object of interest is average age IN THIS CLASS

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$\rightarrow$ Not a random variable, no E or V

Starting example: average age

If the object of interest is average age among ENS students, $E(a)$

You can take this class as a sample from the student population (imagine it is a random sample, which it is not), and:

$$\tau = \frac{1}{N} \sum_i a_i$$

is an estimator for $E(a)$

$\rightarrow$ It is a random variable because another sample (another class) would have produced a different set of a_i's and a different τ (sampling from the population $\implies$ sampling a τ)

Starting example: average age

If individuals are sampled randomly and independently (selecting a class would not work...)

$$E(\tau) = \frac{1}{N}\sum_i E(a_i) = \frac{1}{N}\sum_i E(a) = E(a)$$

$$V(\tau) = \frac{1}{N^2}\sum_i V(a_i) = \frac{1}{N^2}\sum_i V(a) = \frac{1}{N^2}NV(a) = \frac{V(a)}{N}$$

where $V(a)$ is the population variance of a_i (i.e. the variance among all ENS students)

OLS reminder

OLS

$$y = \alpha + \beta x + \varepsilon$$

(only one variable in x)

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For my purpose, get rid of constant:

$$\begin{aligned} \bar{y} &= \alpha + \beta \bar{x} + \bar{\varepsilon} \\ y - \bar{y} &= \beta(x - \bar{x}) + (\varepsilon - \bar{\varepsilon}) \\ \tilde{y} &= \beta \tilde{x} + \tilde{\varepsilon} \end{aligned}$$

OLS

$$\hat{\beta}_{OLS} = \frac{\tilde{x}'\tilde{y}}{\tilde{x}'\tilde{x}}$$

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Because you have a sample (y, x) from a super population, another sample would provide a different (y, x), thus a different $\hat{\beta}_{OLS}$ to estimate the true population relation β

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(In principle) you have learned that:

$$\begin{aligned} E(\hat{\beta}_{OLS}) &= \beta \\ V(\hat{\beta}_{OLS}) &= \frac{V(\varepsilon)}{\tilde{x}'\tilde{x}} \end{aligned}$$

where this variance reflects the variation in ε and x (equivalently, in y and x) when you draw a new sample.

▸ Details

Design-based uncertainty

In a randomized design,
there is another source of uncertainty
than sampling of the population used in the data.
Which is... ?

Fixed sample

Think of your population as given: there is no super-population, your sample *is* the population of interest (size N) over which you define your treatment effect

Your parameter of interest is well defined as:

$$\tau_{fs} = \frac{1}{N}\sum_i [Y_i(1) - Y_i(0)]$$

which is an **estimand**, not an estimator, because you don't observe both $Y_i(1)$ and $Y_i(0)$
(fs stands for "fixed sample")

Fixed sample

Randomize your population into treatment and control and use:

$$\hat{\tau}_{fs} = \frac{1}{N_1} \sum_{i/T_i=1} Y_i(1) - \frac{1}{N_0} \sum_{i/T_i=0} Y_i(0)$$

That is a random variable, because each possible randomization would generate a different estimation value. It is an unbiased estimator of τ_{fs} under randomization

Exemple

Population with 6 observations

i	$Y_i(0)$	$Y_i(1)$	τ_{fs}	Draw 1	$\hat{\tau}_{fs}$	Draw 2	$\hat{\tau}_{fs}$
1	1	3		1		1	
2	2	5		1		0	
3	2	0		1		1	
4	4	4		0		0	
5	0	3		0		1	
6	1	0		0		0	
			0.83		1.00		-0.33

Design-based inference

The logic for inference becomes completely different: not related to population sampling, but to randomization

And the parameters are different: $\tau_{fs} = \frac{1}{N}\sum_i [Y_i(1) - Y_i(0)]$ vs. $E[Y(1) - Y(0)]$

Question of perspective whether you want to hold population fixed or not

$V(\hat{\tau}_{fs})$?

Because the source of randomness is different, the variance of this estimator will be different from the standard OLS variance that is based on population sampling

The way it works is instructive...

Let's do it

Let's do it

Stay super simple, to understand the logic: $N = 2$.

Estimand:

$$\tau_{fs} = \frac{1}{2}\left[(Y_1(1) - Y_1(0)) + (Y_2(1) - Y_2(0))\right]$$

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Estimand:

$$\tau_{fs} = \frac{1}{2}\left[(Y_1(1) - Y_1(0)) + (Y_2(1) - Y_2(0))\right]$$

One is treated, the other is control: $T_1 = 1 - T_2$ (with proba $1/2$)

$$\hat{\tau}_{fs} = T_1(Y_1(1) - Y_2(0)) + (1 - T_1)(Y_2(1) - Y_1(0))$$

Trick

Let's define

$$D = 2T_1 - 1$$

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And

$$\begin{aligned} E(D) &= E(2T_1 - 1) = 2E(T_1) - 1 = 0 \\ V(D) &= E(D^2) - E(D)^2 = E(D^2) = 1 \end{aligned}$$

because $D = 1$ or $D = -1$ (two states of nature)

A bit of manipulation

$$\hat{\tau}_{fs} = T_1(Y_1(1) - Y_2(0)) + (1 - T_1)(Y_2(1) - Y_1(0))$$

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$$
\begin{aligned}
\hat{\tau}_{fs} &= T_1(Y_1(1) - Y_2(0)) + (1 - T_1)(Y_2(1) - Y_1(0)) \\
&= \frac{(1 + D)}{2}(Y_1(1) - Y_2(0)) + \frac{(1 - D)}{2}(Y_2(1) - Y_1(0))
\end{aligned}
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&= \frac{1}{2}\left[(Y_1(1) - Y_2(0)) + (Y_2(1) - Y_1(0))\right] \\
&\quad + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]
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&\quad + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right] \\
&= \tau_{fs} + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]
\end{aligned}$$

Important

$$\hat{\tau}_{fs} = \tau_{fs} + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]$$

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AND

$Y_1(1)$, $Y_2(1)$, etc., are not random: they are given characteristics of a given population

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$$[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))]$$

drives the error over the estimator, and is positive if $D = 1$ and negative if $D = -1$ (you have only one combination of treated vs control, whereas the truth averages the two combinations).

Bias and variance

As D is the only random variable here,

$$\hat{\tau}_{fs} \quad = \quad \tau_{fs} + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]$$

Bias and variance

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\begin{aligned}
\hat{\tau}_{fs} &= \tau_{fs} + \frac{D}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right] \\
E(\hat{\tau}_{fs}) &= \tau_{fs} + \frac{E(D)}{2}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right] = \tau_{fs}
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V(\hat{\tau}_{fs}) &= \frac{V(D)}{4}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]^2
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V(\hat{\tau}_{fs}) &= \frac{V(D)}{4}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]^2 \\
&= \frac{1}{4}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]^2
\end{aligned}$$

"Infeasible" variance

$$V(\hat{\tau}_{fs}) = \frac{1}{4}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]^2$$

We say that this variance is "infeasible" in the sense that it can never be estimated

Why is it impossible to estimate

$$V(\hat{\tau}_{fs}) = \frac{1}{4}\left[(Y_1(1) - Y_2(0)) - (Y_2(1) - Y_1(0))\right]^2$$

?

General result

For $N > 2$, define:

$$
\begin{aligned}
S_0^2 &= \frac{1}{N-1}\sum_i \left[(Y_i(0) - \bar{Y}(0)\right]^2 \\
S_1^2 &= \frac{1}{N-1}\sum_i \left[(Y_i(1) - \bar{Y}(1)\right]^2
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Do you recognize those objects? Are they estimable?

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Do you recognize those objects? Are they estimable?

$$S_{01}^2 = \frac{1}{N-1}\sum_i [(Y_i(1) - Y_i(0) - \tau_{fs}]^2$$

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What is this, is this estimable?

Variance of the effect

$$S_{01}^2 = \frac{1}{N-1} \sum_i \left[(Y_i(1) - Y_i(0) - \tau_{fs} \right]^2$$

Is the variance of the effect (proof not difficult, but tedious)

It cannot be estimated, because it would require to measure the effect individual by individual and then take the variance, but we can't

(Equivalently, we would need the covariance of $(Y(1), Y(0))$ which is not observable).

Unfortunately...

$$V(\hat{\tau}_{fs}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1} - \frac{S_{01}^2}{N}$$

Which is infeasible

Intuitions

$$V(\hat{\tau}_{fs}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1} - \frac{S_{01}^2}{N}$$

Why would the variance of the estimator increase with S_0^2 or S_1^2 ?

Intuition

$$V(\hat{\tau}_{fs}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1} - \frac{S_{01}^2}{N}$$

The intuition for why more precision when $\frac{S_{01}^2}{N}$ is large (treatment effect is more heterogenous) is not straightforward ▸ Intuition

Neyman estimator

Instead of

$$V(\hat{\tau}_{fs}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1} - \frac{S_{01}^2}{N}$$

if we use

$$V^{Ney}(\hat{\tau}_{fs}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1}$$

our inference will be conservative (we overestimate the variance): not optimal but safe. And no other choice.

Design-based and Sampling-based uncertainty

Another Variance estimator

If you take the perspective that your experimental sample is randomly drawn from a super population AND treatment is allocated randomly, you have the two sources of uncertainty

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If you take the perspective that your experimental sample is randomly drawn from a super population AND treatment is allocated randomly, you have the two sources of uncertainty

They BOTH contribute the the variance of the estimator

Another Variance estimator

If you take the perspective that your experimental sample is randomly drawn from a super population AND treatment is allocated randomly, you have the two sources of uncertainty

They BOTH contribute the the variance of the estimator

If you compute the variance taking into account both sources

$$V(\hat{\tau}) = \frac{S_0^2}{N_0} + \frac{S_1^2}{N_1}$$

(Be careful, the estimand is now $\tau = E(Y(1) - Y(0))$, average effect in the infinite super population)

Abadie et al., Econometrica 2020

What is nice?

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So, in a sense, that lecture has no practical implications.

Was it useless? You may think yes; I think no. You should now understand better why an estimator is a random variable, and that there are interesting conceptual issues around that notion.

Have a nice day

OLS variance

Start with the variance conditional on $\tilde{x}$:

$$V(\hat{\beta}_{OLS}|\tilde{x}) \quad = \quad V(\beta) + V\left(\frac{\tilde{x}\varepsilon}{\tilde{x}'\tilde{x}}|\tilde{x}\right)$$

OLS variance

Start with the variance conditional on $\tilde{x}$:

$$
\begin{aligned}
V(\hat{\beta}_{OLS}|\tilde{x}) &= V(\beta) + V\left(\frac{\tilde{x}\varepsilon}{\tilde{x}'\tilde{x}}|\tilde{x}\right) \\
&= \frac{\tilde{x}' V(\tilde{\varepsilon})\tilde{x}}{(\tilde{x}'\tilde{x})^2} \\
&= \frac{V(\tilde{\varepsilon})}{\tilde{x}'\tilde{x}}
\end{aligned}
$$

Using that ε is distributed independently from $\tilde{x}$ (and $\tilde{x}'\tilde{x}$ is a scalar here). We also need it to be homoskedastic.

OLS variance

Now if you want $V(\hat{\beta}_{OLS})$ you need to also include the uncertainty over $\tilde{x}$

Use variance decomposition:

$$V(\hat{\beta}_{OLS}) = E_x[V(\hat{\beta}_{OLS}|\tilde{x})] + V_x[E(\hat{\beta}_{OLS}|\tilde{x})]$$

OLS variance

Now if you want $V(\hat{\beta}_{OLS})$ you need to also include the uncertainty over $\tilde{x}$

Use variance decomposition:

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\begin{aligned}
V(\hat{\beta}_{OLS}) &= E_x[V(\hat{\beta}_{OLS}|\tilde{x})] + V_x[E(\hat{\beta}_{OLS}|\tilde{x})] \\
&= E_x[V(\hat{\beta}_{OLS}|\tilde{x})] + V_x[\beta] \\
&= E_x\left[\frac{V(\tilde{\varepsilon})}{\tilde{x}'\tilde{x}}\right] \\
&= V(\tilde{\varepsilon})E_x[\frac{1}{\tilde{x}'\tilde{x}}]
\end{aligned}
$$

where in practice the best estimator you have for $E_x[1/(\tilde{x}'\tilde{x})]$ is $1/(\tilde{x}'\tilde{x})$

▸ Back

Intuition for variance component

If treatment effect is constant, assume $Y_2(0)$ is small and $Y_1(0)$ is large; so $Y_2(1)$ is small and $Y_1(1)$ is large, to the same extent. Then, if 1 is treated, $Y_1 - Y_2 = Y_1(1) - Y_2(0)$ is as large as it can be; and inversely if 2 is treated, $Y_2 - Y_1 = Y_2(1) - Y_1(0)$ is as small as it can be. So there is a lot of variance in the **estimated** effect (but not in the true effect, which is why this is not intuitive !).

▸ Back

OLS variance is the Neyman estimator

Consider $\tilde{Y} = \beta \tilde{T} + \tilde{\varepsilon}$. Treat $\tilde{T}$ just as $\tilde{x}$ above. Notice that $\bar{T} = N_1/N$. Then $\tilde{T}'\tilde{T} = (N_1 N_0)/N$

And OLS variance is:

$$\frac{V(\tilde{\varepsilon})N}{N_1 N_0}$$

OLS variance is the Neyman estimator

Now start with a simple case: variance of $Y(1)$ is the same as variance of $Y(0)$, so $S_0^2 = S_1^2$. Notice also that $V(Y(0)) = V(\tilde{\varepsilon})$

Neyman variance is $S_0^2/N_0 + S_0^2/N_1 = S_0^2 N/(N_1 N_0)$ which is also the OLS variance estimator above.

OLS variance is the Neyman estimator

To recover the Neyman estimator in the general case, you need to be familiar with the Huber-White robust variance in OLS. Assuming you are, it goes like this: $V(\tilde{\varepsilon}) = \Omega$ and OLS robust variance is

$$(\tilde{T}'\tilde{T})^{-1}(\tilde{T}'\Omega\tilde{T})(\tilde{T}'\tilde{T})^{-1}$$

where

$$(\tilde{T}'\Omega\tilde{T})$$

is estimated by $\sum_i \tilde{T}_i^2 \hat{\tilde{\varepsilon}}_i^2 = \sum_{i/T_i=1}(1-\bar{T})^2 \hat{\tilde{\varepsilon}}_i^2 + \sum_{i/T_i=0} \bar{T}^2 \hat{\tilde{\varepsilon}}_i^2$. Using $(\tilde{T}'\tilde{T})^{-1} = N/(N_1 N_0)$, a scalar, boils down to the Huber-White variance being

$$\sum_{i/T_i=1} \frac{\hat{\tilde{\varepsilon}}_i^2}{N_1} + \sum_{i/T_i=0} \frac{\hat{\tilde{\varepsilon}}_i^2}{N_0}$$

which is an estimator of the Neyman variance. Back